FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – III
PAPER-2

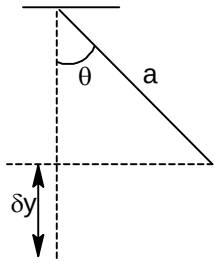
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A, C**	21.	**B**	41.	**A, B, C, D**
2.	**C, D**	22.	**A**	42.	**A, B, C, D**
3.	**B, D**	23.	**B, C**	43.	**A, B, C**
4.	**A, D**	24.	**A, B, C**	44.	**A, B, C**
5.	**B, C**	25.	**B, C, D**	45.	**A, B, C**
6.	**A, C**	26.	**B, C, D**	46.	**A, B, C, D**
7.	**A, C**	27.	**C**	47.	**A**
8.	**A, B, D**	28.	**A, B, D**	48.	**A, B, C**
9.	**C**	29.	**B**	49.	**C**
10.	**C**	30.	**D**	50.	**C**
11.	**B**	31.	**A**	51.	**B**
12.	**A**	32.	**C**	52.	**C**
13.	**5**	33.	**1**	53.	**4**
14.	**1**	34.	**8**	54.	**1**
15.	**2**	35.	**5**	55.	**1**
16.	**8**	36.	**9**	56.	**2**
17.	**2**	37.	**3**	57.	**3**
18.	**5**	38.	**9**	58.	**3**
19.	**3**	39.	**6**	59.	**7**
20.	**4**	40.	**7**	60.	**4**

Physics

PART – I

SECTION – A

1. $\delta y = a(1-\cos\theta) = \frac{a\theta^2}{2}$



and $\theta_a = \frac{b\phi}{2}$

$\Rightarrow \quad \delta y = \frac{a}{2}\left(\frac{b\phi}{2a}\right)^2 = \frac{b^2}{8a}\phi^2$

$\Rightarrow \quad \delta y = \frac{b^2}{8a}(\omega\delta t)^2$

acc. of rod's centre = $\frac{b^2}{4a}\omega^2$

$\Rightarrow \quad \delta F = \frac{mb^2}{4a}\omega^2$

2. $y = a_0 \sin(kx) \sin(\omega t + \phi)$

$\frac{kL}{3} = n\pi$ and $\frac{kL}{2} = \left(n+\frac{1}{2}\right)\pi$

$\Rightarrow \quad n = 1,\ k = \frac{3\pi}{L}$

Also; $a' = \left|a_0 \sin\left(\frac{3\pi}{L}\times\frac{5L}{6}\right)\right| = a_0$

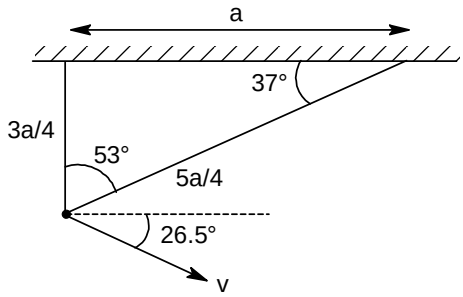
3. $h = \frac{3a}{4}$

$v_0 = \sqrt{2gh}$

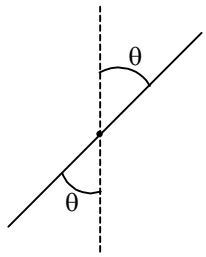
$v_0 = \sqrt{1.5ag}$ and $v = v_0\cos(90°-26.5°)$

$= v_0 \sin(26.5°)$

$\Rightarrow \quad v = \sqrt{1.5ag}\,\sin(26.5°)$



4. Let θ be the angle that the rod rotates about its central axis and ϕ be the angle made by string with the vertical.



$-(2T\phi)\ \frac{L}{2} = \frac{mL^2}{12}\frac{d^2\theta}{dt^2}$

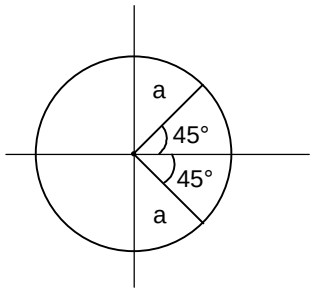
$\Rightarrow \quad \frac{d^2\theta}{dt^2} = -\left(\frac{6g}{L}\right)\phi$

or $\frac{d^2\theta}{dt^2} = -\left(\frac{3g}{\ell}\right)\theta$

$\Rightarrow \quad T_2 = 2\pi\sqrt{\frac{\ell}{3g}}$

5. $\frac{-GMm}{2R} = \frac{-GMm}{R} + \frac{1}{2}mv_0^2$

$\Rightarrow \quad v_0 = \sqrt{\frac{GM}{R}}$



While in the tunnel;

$\frac{d^2x}{dt^2} = -\left(\frac{GM}{R^3}\right)x$

$v_0^2 = \frac{GM}{R} = \omega^2(a^2 - R^2) \Rightarrow a = \sqrt{2}R$

$\Rightarrow \quad T_{A\to B} = \frac{1}{4} \times 2\pi\sqrt{\frac{R^3}{GM}}$

6. $t^2 = \frac{2\ell}{g[\cos\alpha \sin\alpha - \mu\cos^2\alpha]}$

For min. time; $\frac{d}{d\alpha}(\cos\alpha\sin\alpha - \mu\cos^2\alpha) = 0$

$\Rightarrow \quad \tan 2\alpha = \frac{-1}{\mu}$

$\Delta K + \Delta U = W_{friction}$

$\Rightarrow \quad v_0 = \sqrt{2g\ell(\tan\theta_0 - \mu)}$

7. $v \propto \frac{Z}{n}$, $R \propto \frac{n^2}{Z}$ and $E \propto \frac{Z^2}{n^2}$

$\Rightarrow \quad vR \propto n$ and $\frac{v}{E} \propto n$

8. $mv_0 = (4m + m)v$

$v = \frac{v_0}{5}$

$\frac{1}{2}mv_0^2 = \frac{1}{2}.(5m).\frac{v_0^2}{25} + \Delta E$

$\frac{1}{2}mv_0^2 - \frac{1}{2}.\frac{v^2}{25}m = \Delta E$

$\Rightarrow \quad \frac{1}{2}mv_0^2\left\{\frac{4}{5}\right\} = \Delta E$

$\frac{1}{2}mv_0^2 = \frac{5}{4}\Delta E$

$\Delta E = 13.6$

$\Delta E = 13.6\left\{1 - \frac{1}{4}\right\} = \frac{3}{4}.(13.6)$

$\frac{1}{2}mv_0^2 = \frac{5}{4} \times 13.6, \frac{3}{5} \times 13.6$

= 17 eV, 8.16 eV

9. $mg - T = ma_1$
$T\cos 30° = ma_2$
and $a_2 \cos 30° = a_1$
$\Rightarrow mg = ma_1\left(1+\frac{1}{\cos^2 30°}\right)$
$\Rightarrow \quad a_1 = \frac{3g}{7}$

10. Height dropped by A
$h = \frac{a}{\cos 60°} - \frac{a}{\cos 37°} = \frac{3a}{4}$
$mgh = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2$
Also, $v_B \sin 37° = v_A$
$\Rightarrow \frac{3ag}{4} = \frac{v_A^2}{2}\left[1+\left(\frac{5}{2}\right)^2\right]$
$v_A = \frac{3}{2}\sqrt{\frac{3ag}{17}}$

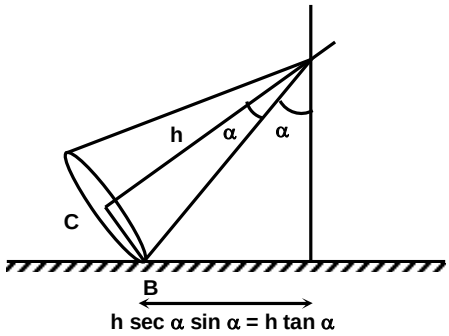
11. $P_{lower} - P_{upper} = P_0$
$\Rightarrow \frac{3P_0 \times (8)}{(8+x)} - \frac{2P_0 \times 12}{(28-x)} = P_0$
$\Rightarrow x = 4$ cm

12. $W_{total} = W_{upper} + W_{lower}$
$= 300R\left[\ell n\left(\frac{24}{12}\right) + \ell n\left(\frac{12}{8}\right)\right] = 300R\ell n(3)$

SECTION – C

13. $F_A = \frac{2m}{3}\omega^2 \times \frac{2\ell}{3}$, $F_C = \frac{m}{2}\omega^2 \times \frac{3\ell}{4}$
$\Rightarrow \frac{F_A}{F_C} = \frac{32}{27}$
$\Rightarrow n = 5$

14. $\omega_1 h\tan\alpha = \omega_2 R$
$\Rightarrow \frac{\omega_1}{\omega_2} = \frac{R/h}{\tan\alpha} = 1$



15. $I_0 = \dfrac{k_0 A}{\cos\left(\dfrac{\pi}{6L}x\right)}\dfrac{dT}{dx}$

$\Rightarrow T = T_0 + \dfrac{6I_0 L}{\pi k_0 A}\sin\left(\dfrac{\pi}{6L}x\right)$

$\Rightarrow T = 2T_0$ at $x = L$

16. $j = \dfrac{1}{\rho}E = \dfrac{\sigma}{2\rho\varepsilon}$

$I = jA = \dfrac{q}{2\rho\varepsilon} \Rightarrow \dfrac{dq}{dt} = -\dfrac{q}{2\rho\varepsilon}$

$\Rightarrow \quad q = Qe^{-\left(\frac{t}{2\rho\varepsilon}\right)}$

Or $q = \dfrac{Q}{8}$ at $t = 6\rho\varepsilon\,\ell n(2)$

17. $\vec{v} = \dfrac{v_0}{2}\left(\sqrt{3}\hat{i} + \hat{j}\right)$

For crossing origin,

$k\dfrac{2\pi m}{qB} = \dfrac{2(v_0/2)}{qE/m}$; k is an integer

$\Rightarrow B = \left(\dfrac{2\pi E}{v_0}\right)k$

For minimum value; k = 1

$\Rightarrow B_{min} = \dfrac{2\pi E}{v_0}$

18. For leaving the surface, $\cos\theta = 2/3$

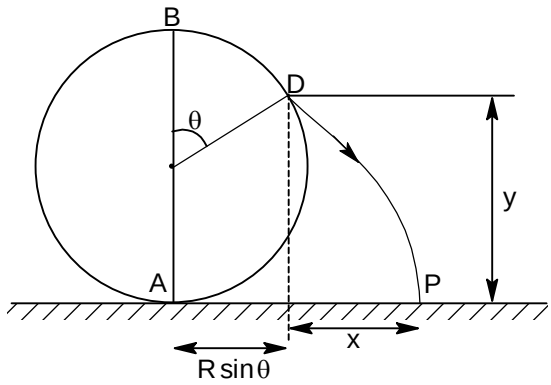
$y = (v\sin\theta)t + \frac{1}{2}gt^2 = R(1+\cos\theta)$ … (1)

and $x = (v\cos\theta)t$ … (2)

and $v^2 = \dfrac{2}{3}gR$ … (3)

On solving;

$AP = R\sin\theta + x = \dfrac{5}{27}(\sqrt{5} + 4\sqrt{2})R$

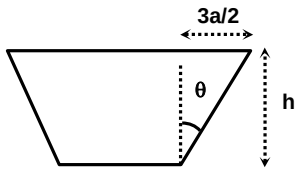


19. $\tan\theta = \dfrac{3a}{2h}$

$2 \times 4a\,T \times \cos\theta = 4a\lambda g$

$\cos\theta = \dfrac{\lambda g}{2T}$

$\Rightarrow h = \dfrac{3\lambda g a}{2\sqrt{4T^2 - \lambda^2 g^2}}$



20. $-A\rho g\, x\,(1+\sin 30^\circ) = A\rho R\left[\frac{\pi}{2}+\operatorname{cosec}30^\circ\right]\frac{d^2x}{dt^2}$

$$\Rightarrow \frac{d^2x}{dt^2} = -\left(\frac{gx}{R}\right)\left(\frac{3/2}{\frac{\pi}{2}+2}\right)$$

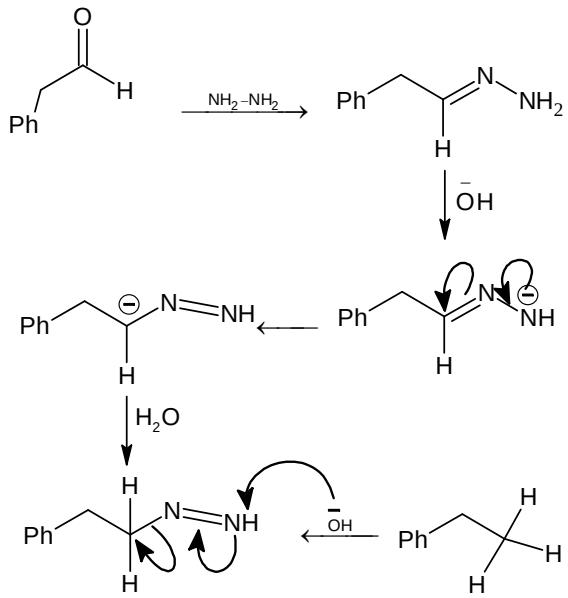
$$\Rightarrow T = \pi\sqrt{\frac{4R(\pi+4)}{3g}}$$

Chemistry

PART – II

SECTION – A

21.



22. Initial m. mole of Cu^{2+} = 5
m. mole of H^+ produced = 1
m. mole of Cu^{2+} converted to Cu = $\frac{1}{2} = 0.5$
m. mole of Cu^{2+} remaining in solution = 5 – 0.5 = 4.5

$$2Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 + I_2$$

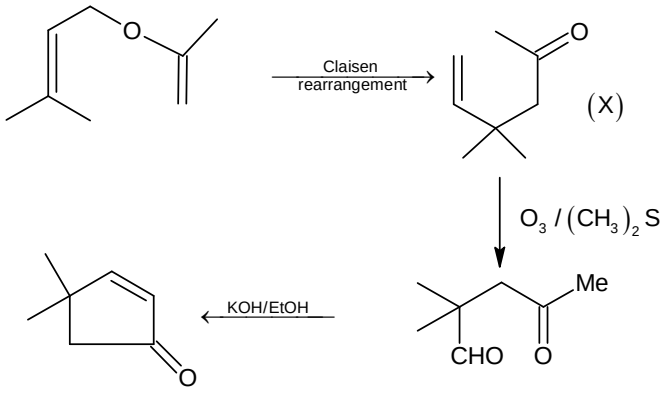
$$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$$

4.5 = 0.04 × V
V = 112.5 mol

23. $\left.\begin{matrix} B \longrightarrow 2R,3S \\ C \longrightarrow 2R,3S \end{matrix}\right\}$ Both are meso.

24. Only (D) option is correct. π electron cloud of $C_2 - C_3$ is present in same plane.

25.



26. $2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

$Br_2 + 2\bar{O}H \longrightarrow BrO^- + Br^- + H_2O$

$Cl_2 + SeO_3^{2-} + H_2O \longrightarrow SeO_4^{2-} + 2Cl^- + 2H^+$

$Cl_2 + IO_3^- + 2OH^- \longrightarrow 2Cl^- + IO_4^- + H_2O$

27. $(COOH)_2 \xrightarrow{\Delta} \underset{X}{CO_2} + \underset{Y}{CO} + \underset{Z}{H_2O}$

CO_2 → non polar and acidic → sp
CO → polar and neutral → sp
H_2O → polar, pH = 7 → sp^3

28. $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

$(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} N_2 + + Cr_2O_3(s) + 4H_2O$

$NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$

Solution for the Q. No. 29 to 30.

$X_2 = O_2$, $X_3 = O_3$, $Y_2 = I_2$

$2I^- + H_2O + O_3 \longrightarrow 2OH^- + I_2 + O_2$

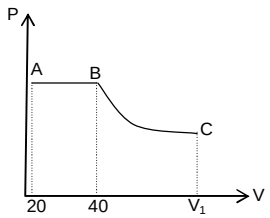
$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

I_2 + Starch solution ⟶ Blue colouration

$I_2 + I^-(\text{excess}) \longrightarrow$ Brown coloured I_3^- ion

O_2 is thermodynamically more stable as compared to O_3
O_3 is pale blue gas which has characteristic smell.

32.



A → P_1, 300 K
B → P_1, T_2
C → P_2, 300 K

For state A $P_1 = \dfrac{nRT}{V} = 2.46$ atm

For isobaric process AB $\dfrac{V_1}{T_1} = \dfrac{V_2}{T_2}$

$T_2 = 600$ K

For adiabatic process BC $TV^{\gamma-1} = \text{Constant}$

$600(40)^{\gamma-1} = 300(V_1)^{\gamma-1}$

$V_1 = 113.13$ lit.

SECTION – C

33. $Mg^{2+} + \underset{\text{(Hexadentate ligand)}}{EDTA^{4-}} \longrightarrow \underset{\text{Octahedral complex}}{\left[Mg(EDTA)\right]^{2-}}$ CN = 6

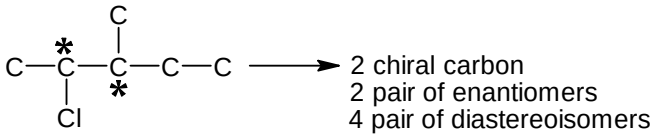
34. $PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

35. H_2, Li_2, B_2, N_2, F_2

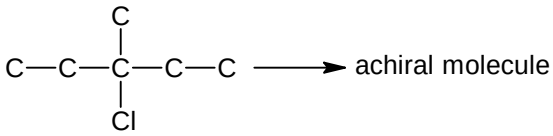
36. Atomic number of metal = 29
$Cu = 3d^{10}4s^1$
$Cu^{2+} = 3d^9$

37. $2KI + HgI_2 \longrightarrow K_2[HgI_4] \longrightarrow 2K^+ + [HgI_4]^{2-}$
i = 3

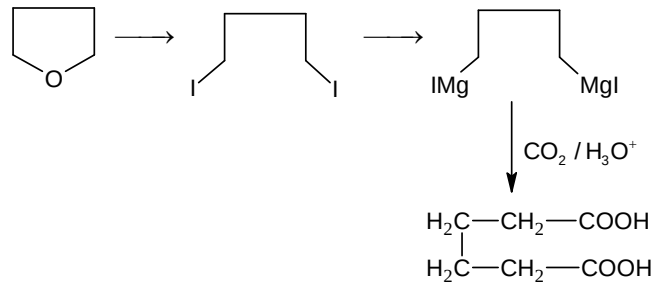
38. $Cl-C-C-\overset{\overset{C}{|}}{\underset{*}{C}}-C-C \rightarrow$ one chiral carbon pair of enantiomers.







x = 3, y = 4, z = 2
x + y + z = 9

39. 

40. $\because K_{sp}(CuCl) >> K_{sp}(AgCl)$
Thus assume $[Cl^-]$ in solution comes from CuCl only.
$[Cl^-] = \sqrt{K_{sp}CuCl} = 10^{-3}$ M
For AgCl $[Ag^+][Cl^-] = 1.6 \times 10^{-10}$
$[Ag^+] = 1.6 \times 10^{-7}$
x = 7

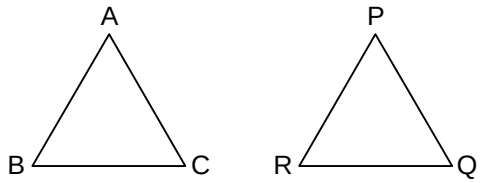
Mathematics

PART – III

SECTION – A

41. $\frac{d}{dx}\left(\left(f(x)\right)^2+\left(f'(x)\right)^2\right)=2f'(x)\cdot f(x)+2f'(x)\cdot f''(x)=-2x\ g(x)(f'(x))^2\le 0$

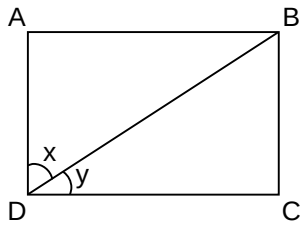
42. $|PR|^2+|RQ|^2\ge 2|PR|\,|RQ|$
$|PR|\cdot|RQ|\ge 2Ar(\Delta PQR)$
$\Rightarrow 8Ar(\Delta PQR)\le |PR|^2+|RQ|^2+4Ar(\Delta PQR)$
$\le |PR|^2+|RQ|^2+2\cdot|PR|\cdot|RQ|<8Ar(\Delta PQR)+1$
$\Rightarrow 8Ar(\Delta PQR)=|PQ|^2+|QR|^2+4Ar(\Delta PQR)$ and
$|PQ|^2+|QR|^2=2|PQ|\cdot|QR|=4Ar(\Delta PQR)$
$\therefore\ \angle R=90^\circ$ and $RP=RQ$



43. $\vec{p}+\vec{q}+\vec{r}=0$
So, $\vec{p},\vec{q},\vec{r}$ can form a triangle
$\vec{p}\cdot\vec{c}=\left(\vec{b}\cdot\vec{c}\right)\left(\vec{a}\cdot\vec{c}\right)-\left(\vec{c}\cdot\vec{a}\right)\left(\vec{b}\cdot\vec{c}\right)=0$
$\therefore\ \vec{p}\perp\vec{c}$
$\Rightarrow\ \vec{q}\perp\vec{a}$ and $\vec{r}\perp\vec{b}$

44. Take, $\beta=1$
$\Rightarrow |1+\alpha|^2=|(1+\cos\theta)+i\sin\theta|^2=2+2\cos\theta\ge 2+\sqrt{3}$
$\Rightarrow\ \cos\theta\ge\frac{\sqrt{3}}{2}$ so, we get 332 angles
$\therefore\ p=\frac{332}{1996}$

45. $2x=\angle CBD$
$2y=\angle ABD$
In ΔCBD
$\frac{\sin\left(\pi-(2y+x)\right)}{\sin x}=\frac{BD}{BA}=\frac{BD}{BC}=\frac{\sin\left(\pi-(2x+y)\right)}{\sin y}$
$\Rightarrow \sin(2y+x)\sin y=\sin(2x+y)\sin x$
$\Rightarrow\ \frac{1}{2}\left[\cos(y+x)-\cos(3y+x)\right]=\frac{1}{2}\left[\cos(x+y)-\cos(3x+y)\right]$
$0<x+y=\frac{1}{2}ABC<\frac{\pi}{2}$
$0<(3y+x)+(3x+y)<2\pi$
$\therefore\ 3y+x=3x+y\Rightarrow x=y$
$\Rightarrow\ \angle ABD=\angle CBD\Rightarrow AD=CD$



46. $g(x)=f(x)\cdot f\left(\frac{a}{x}\right)$
$g'(x)=f'(x)\cdot f\left(\frac{a}{x}\right)-f(x)f'\left(\frac{a}{x}\right)\left(\frac{a}{x^2}\right)=\frac{a}{x}-x\left(\frac{a}{x^2}\right)=0$
$\Rightarrow g(x)=b$

$\Rightarrow f(x)\cdot f\left(\frac{a}{x}\right)=b$

$\Rightarrow f(x)=\sqrt{b}\left(\frac{x}{\sqrt{a}}\right)^{\frac{a}{b}}$

47. Let, OA = x_1, OB = y_1, OC = z_1 be points on X, Y, Z axis then equation of plane is

$\frac{x}{x_1}+\frac{y}{y_1}+\frac{z}{z_1}=1$

$\Rightarrow \frac{1}{x_1}+\frac{2}{y_1}+\frac{3}{z_1}=1$

$\therefore$ Locus is $\frac{1}{x}+\frac{2}{y}+\frac{3}{z}=1$

48. Total ways = $^8C_3+{}^8C_4\times{}^4C_2+{}^8C_2\times\frac{^6C_3}{2}+{}^8C_3\times{}^5C_1\times\frac{^4C_2}{2}+\frac{\lfloor\underline{8}}{(2)^4\lfloor\underline{4}}$

49.-50. $f(x^2 + y\,f(z)) = x\cdot f(x) + z\,f(y)$

$x = y = z = 0$

$f(0) = 0$

$x = 0$

$f(y\cdot f(z)) = z\cdot f(y)$

$y = z = t$

$f(t\cdot f(t)) = t\cdot f(t)$

Using this and substitution, we get $f(x) = x$ and $f(x) = 0$

51.-52. Ar.$(B_0, B_1, \ldots, B_{n-1}) = \frac{n}{4}\cot\left(\frac{\pi}{n}\right)$

Ar.$(A_0 A_1 \ldots A_{n-1}) = \frac{n}{2}\sin\frac{2\pi}{n}$

$\therefore \frac{\text{Ar.}(A_0\ A_1 \ldots A_n)}{\text{Ar.}(B_0\ B_1 \ldots B_n)}=\frac{n\sin\frac{\pi}{n}\cos\frac{\pi}{n}}{\frac{n}{4}\cot\frac{\pi}{n}}=4\sin^2\frac{\pi}{n}$

SECTION – C

53. $x(f(x))^2 - x^2f(x) = \left(\sqrt{x}f(x)\right)^2 - 2\cdot\sqrt{x}\times f(x)\cdot\frac{x^{3/2}}{2}+\frac{x^3}{4}-\frac{x^3}{4}$

$\left(\sqrt{x}\,f(x)-\frac{x^{3/2}}{2}\right)^2-\frac{x^3}{4}$

$\therefore B - A = \int_0^1\left(\sqrt{x}\,f(x)-\frac{x^{3/2}}{2}\right)^2 dx-\int_0^1\frac{x^3}{4}dx$

$\therefore A - B \le \frac{1}{16}$

54. Say, $\lambda < 0$

$\omega = \sqrt{-\lambda}$
$\therefore |I + \lambda A^2| = |I - \omega^2 A^2| = |I - \omega A||I + \omega A|$
As, $A = -A'$
$I - \omega A = I + \omega A' = (I + \omega A)' \Rightarrow |I + \lambda A^2| = |I + \omega A||I + \omega A|' = |I + \omega A|^2 \geq 0$
Same, can be seen for $\lambda \geq 0$

55. If $x > 2$ then $x^3 - 3x > 4x - x = x > \sqrt{x+2}$
$\Rightarrow |x| \leq 2$ take $x = 2\cos\theta$ for some $\theta \in [0, \pi]$
$\Rightarrow 2\cos 3\theta = \sqrt{2(1+\cos\theta)}$
$\Rightarrow 2\sin\frac{7\theta}{4} \cdot \sin\frac{5\theta}{4} = 0$
$\Rightarrow \theta = 0, \frac{4\pi}{7}, \frac{4\pi}{5}$
$\therefore x = 2, 2\cos\frac{4\pi}{7}, -\frac{1}{2}\left(1+\sqrt{5}\right)$

56. $B^2 - tr(B)\cdot B + I = 0$
$\Rightarrow AB - (tr(B)A + AB^{-1} = 0$
$\therefore tr(AB) - tr(A)tr(B) + tr(AB^{-1}) = 0$

57. Using property

58. Using property of parabola

59. $\lim\limits_{x\to\infty} \frac{f(x) - f(\alpha x)}{x} = 0$
$\Rightarrow$ For any $\in > 0$ there is $\delta > 0$ such that $|x| < \delta$ and $|f(x) - f(\alpha x)| < \in |x|$ using triangle inequality
$|f(x) - f(\alpha^n x) \leq |f(x) - f(\alpha x)| + |f(\alpha x) - f(\alpha^2 x)| + ... + |f(\alpha^{n-1} x) - f(\alpha^n x)|$
$< \in |x|(1 + \alpha + \alpha^2 + ... + \alpha^{n-1}) = \in \frac{1-\alpha^n}{1-\alpha}|x| \leq \frac{\in |x|}{1-\alpha}$
As, $n \to \infty$
$|f(x)| \leq \frac{\in}{1-\alpha}|x|$
$\Rightarrow \lim\limits_{x\to\infty} \frac{f(x)}{x} = 0$

60. Let $g(x) = f'(x)e^{-\lambda f(x)}$
$g(a) = g(b)$ so using Rolle's theorem in $C \in (a, b)$
$g'(c) = 0$
$\Rightarrow g'(x) = e^{-\lambda f(x)} (f''(x) - \lambda(f'(x)^2)$
$\Rightarrow f''(c) - \lambda(f'(c))^2 = 0$